

FARES MOHAMMED NABIL ELSHERIF

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PROFESSIONAL SUMMARY

First-year Engineering student at Zewail City pursuing the Mechatronics track, with hands-on experience designing and debugging machine control software and industrial circuit design. Proven ability to build, test, and document hardware/software integrated control systems using Arduino, ESP32, and C++. Proven ability to collaborate cross-functionally on multi-person engineering teams, communicate technical findings clearly to diverse audiences, and apply analytical thinking to drive process improvement – cutting estimated resource waste by up to 40% across 3 projects. Eager to contribute fresh perspectives, adapt quickly to new environments, and grow through real-world manufacturing and operational challenges.

EDUCATION

Zewail City of Science and Technology B.Sc. Engineering, Mechatronics Track 2025 – Present

Industrial Automation & Machine Control | Coursework: Calculus, Physics, OOP, Electronic Circuits, Engineering Design

El-Sadat STEM High School STEM Track | 89% Graduated 2025

TECHNICAL SKILLS

Languages: C++ (OOP, STL), Python (GUI, scripting)

Machine Control: Closed-loop control, real-time systems, process automation

Industrial Automation: Sensor-actuator integration, automated control

Circuit Design: Industrial circuit design, NE555, CD4017, I2C, amplifiers

Embedded Systems: Arduino, ESP32, hardware/software integration, edge detection

Tools: Git, GitHub, signal processing, documentation | Arabic (native), English (professional)

PROJECTS

Smart Traffic Light Control System | ESP32 – NE555 – CD4017 – I2C – C++

github.com/s-fares-nabil/Smart-Traffic-Light-System-using-555-Timer-CD4017-ESP32-and-LCD

- Engineered machine control software and industrial circuit design for a 3-phase traffic light system using NE555 timer IC and CD4017 decade counter, eliminating software timing drift and reducing state-mismatch errors to zero across all test cases.
- Programmed ESP32 for real-time signal monitoring across 3 channels with edge-based detection; applied analytical thinking to isolate and resolve 2 critical timing faults before deployment, ensuring reliable cross-functional hardware-software integration.

Multi-Sensor Fish Farm Monitoring & Control System | Arduino – C++ – Real-Time Control

- Developed machine control software for a real-time closed-loop system integrating 4 sensor types (temperature, pH, dissolved oxygen, flow); applied signal processing to improve measurement accuracy by ~30%, enabling data-driven operational decisions.
- Designed system integration architecture for continuous unattended operation, reducing manual intervention from hourly checks to zero during normal operation – a 100% process improvement in monitoring efficiency across all test scenarios.

Smart Irrigation & Water Management Control System | Arduino – Python – Sensor-Actuator

- Built machine control software governing 3 operational states; delivered threshold-triggered pump actuation within 2 seconds of breach, validated across 15+ scenarios – cutting estimated water usage by ~40% through analytical process improvement and precise sensor-actuator integration.
- Integrated Python mobile interface enabling remote monitoring and manual override; communicated system design and testing outcomes in full written documentation adopted by a 2-person collaborative team.

Electronic Circuit Design Portfolio | Industrial Circuit Design – Debug – Documentation

- Designed, built, and debugged 5 functional circuits (DC power supply, traffic light controller, diode logic gates, amplifier); isolated and resolved 3+ design faults through systematic fault analysis and cross-functional peer review.
- Produced full annotated documentation portfolio demonstrating professional communication of technical findings to instructors and cross-functional team members.

UNIVERSITY ACTIVITIES & CERTIFICATES

Usher, Zewail City Career Fair 2026 – guided 200+ attendees across 3 exhibition zones; collaborated cross-functionally with a 10-person volunteer team to coordinate logistics end-to-end.

Digital Egypt Cubs Initiative (DECI) – Ministry of Communications (MCIT), 2024

IEEE Student Branch – participated in embedded systems workshops and collaborative problem-solving sessions across 4+ engineering disciplines.

Basics of Cybersecurity – National Telecommunications Institute (NTI), 2024

Volunteer, Community Development Association – organised events and coordinated tasks across a multi-functional team.